

Martin Chow

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Skills

Rendering: Vulkan, real-time ray tracing, PBR, Forward+ / deferred pipelines, GPU-driven rendering

Engine: C++ engine architecture, render graph systems, animation, physics integration, editor tools

Performance: Profiling, frame-time analysis, memory optimization, multithreading, PC hardware debugging

Systems: Cross-platform C++, Windows/Linux, resource pipelines, tooling, CI/CD

Tools: Unreal Engine 5, ImGui, CMake, Git, Python, debugging utilities, custom dev tools, Blender

Experience

Microsoft — SQL Performance Team

Redmond, WA

Software Engineer

2023 – 2025

- Analyzed and resolved performance bottlenecks across large-scale distributed systems and developer tools.
- Built diagnostics, profiling utilities, and CI-integrated performance validation pipelines.
- Improved reliability and responsiveness of internal tools used by multiple engineering teams.
- Collaborated with engineers, PMs, and QA to deliver fixes and improvements with player-like quality expectations.

University of Washington — DAIS Lab

Seattle, WA

Research Software Engineer

2022 – 2023

- Developed C++ / Python tools supporting simulation workflows and scientific visualization.
- Built automation pipelines and debugging utilities used by researchers for iterative experimentation.
- Worked closely with cross-disciplinary teams to improve usability, stability, and workflow efficiency.

Projects

C++ Personal Project

LuareEye Engine — Custom 3D Engine

2025

- Designed and implemented a full rendering pipeline (Deferred, GPU-driven) with a modular engine architecture.
- Built editor tools (material editor, spline/path editor), asset importers, and real-time debugging overlays.
- Developed the full animation pipeline, including skeletal animation, blending, animation events, and runtime retargeting.
- Integrated Jolt physics; implemented CCD for high-speed collision stability.
- Improved performance across varied PC hardware, reducing GPU load by 18% and increasing frame rate by 22%.

C++ Personal Project

Vulkan Real-Time Ray Tracing Engine

2024

- Implemented BLAS/TLAS acceleration structures, SBT setup, and ray tracing pipelines.
- Added reflections, soft shadows, GI, and SVGF denoising with TAA accumulation.
- Debugged synchronization issues, memory barriers, and GPU timing stalls across multiple PC configurations.

C++ Personal Project

Offline Path Tracer

2024

- Built a CPU path tracer with MIS, BRDF models, BVH acceleration, and physically-based lighting.

C++ Personal Project

Physics Simulation Engine — Rigid-Body Dynamics

2025

- Implemented rigid-body dynamics, constraint solvers, inertia tensors, and multiple integrators (Euler, Verlet, RK4).
- Created real-time visualization tools for debugging collision geometry and solver behavior.

Education

DigiPen Institute of Technology

Redmond, WA

Master of Science in Computer Science

2024 – 2026 (expected)